Psismos

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Abstract

This report presents our tangible media semester project: Psismos, a music controller artifact. We explored somatics through movement and pressure by using wearable sensors. We were especially inspired by the rave as an environment: a cultural space flooded with sound and movement that can encourage unconventionality, subversion, and release. We explored different design questions, made a small-scale prototype to test our concept, built an interactive synth that is modifiable through sensors and built a wearable that inputs data and outputs sound & haptics.

Design Narrative

Somatics and movement

Somatics and movement served as our anchoring point. The work then branched off into rave culture, drawing from its spiritual, bodily and communal dimensions. As our contemporary culture is heavily involved in self-observation, surveillance and distraction, we aimed to create a narrative about body and mind connection, community and the individual's relationship to sound.

Compression & release

Compression—the increase of pressure—was an interesting interaction from the beginning due to its affective connotations. It finds itself in many metaphors (applying pressure, being under pressure, peer pressure), most of them referring to the application of stress onto an individual or situation. Release, on the other hand, is etymologically related to relaxation. Progressive Muscle Relaxation is a meditation technique that involves tensing one's muscles and relaxing them in succession, creating awareness of bodily tension. One can only release after compressing. This complementary relationship became a foundational tension in our exploration.

Materials exploration

Our materials exploration aimed to extend the project's design values: somatics and movement, compression and release. Materials with elasticity were especially compelling because they allowed for deformation and return to form. We looked into sleeve textiles

that were comfortable to move in, strong enough to hold all the components and sufficiently stretchy. For the compression device, we focused on a form that could be deformed easily, receive measurable pressure, and still return quickly to its original shape.

We knew that we wanted the device to feel anchored to the wearer, so we looked for materials that would mold to the body well and tightly. Across the form exploration, we referenced material elements from different subcultures (mainly rave, bondage and industrial) to convey unconventionality with the goal to provide an environment to facilitate mindfulness, self-expression and a deferral of self-judgment.

Inspiration
Several pieces guided the creative direction we undertook with this project. As we wanted to explore biometric data sonification, the work of artists Erin Gee (the BioSynth) and Atau Tanaka (Om Ujjayi) helped us see how certain signals (ECG, EMG, skin conductance) could be used in a creative context. Similarly, the integration of haptics was informed by sound artist Stefanie Egedy's work with subwoofers as a way to play with the body's somatosensory perception of pressure and vibrations, effectively composing with/for the body through low frequency vibrations. Another project of interest was Spirit System of Soft Knowing by the artist collective Keiken, an installation in which a user can place a haptic womb on their abdomen while wearing headphones to both feel and hear a marine soundscape.

Prototyping

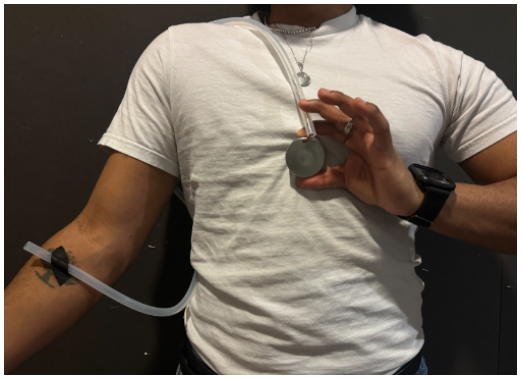
Our first prototype was created on a breadboard with a photoresistor and pressure sensor. At this stage, it was not wearable yet, but we started exploring the relationship to sound and pressure and the photocell, which was meant as a stand-in for a fluctuating biometric datapoint (like an oximeter).

This iteration used a Max/MSP patch comprised of a spectral synth and a noise layer that would alternate in amplitude according to the applied pressure

Here is a video of the prototype in action:
https://youtu.be/qFe5QgwqWRs?si=0kVX7wUxlSoOPUag

The form went through different moodboarded conceptual iterations, many pulled from material explorations around culture, sensoriality and flexibility.

We started with a unifying black stretchy material for the sleeve. This provided necessary flexibility to be worn by different users and applied enough pressure against the forearm, anchoring the user in the experience.



The placement of the transducer on the sternum was chosen because it was a very grounding location as well—close to the heart, which itself applies continual compression and release to sustain our bodies. It is also an ideal position because it is a large area that allows vibrations to be felt across the body via bone conduction. The haptic sensation on our chest recalled loud music environments in which sounds are felt from within the body.

Through this process of iteration, we refined the scope of our project and finally decided to pivot towards sensing motion rather than biosignals as a way to affect the sonic outputs. Certain Digital Music Instruments (DMIs), such as the T Stick developed at McGill University employ accelerometers to a performers movements. In a way, we still use body-generated data, but choose external movement rather than internal processes to capture a person's embodied state.

Final result

Our final artifact, Psismos, is a performance device consisting of a wearable gestural synthesizer, a speaker and a haptic chest piece. The wearable synth allows a performer to move their arm in multiple ways and squeeze the plastic balloon in order to affect the sonic output. They can then appreciate this output both audibly through the speaker and haptically through the chest piece.

The wearable gestural synthesizer is built using a Raspberry Pi Pico 2W microcontroller coded using CircuitPython. All of the audio synthesis occurs directly on the MCU, via CircuitPython's synthio library. Two synth patches were created using wavetable synthesis: one Kick/Bass patch and one Drone patch. Audio effects, such as delays and phasers were implemented using the audiofilters and audiodelays libraries.

To enable gestural control, the piece is equipped with a MPU6050 (triple axis gyroscope + accelerometer) placed on the hand section of the sleeve. Several data points of the MPU are used to control the synth parameters. For example, the gyroscope X and gyroscope Y control which section of the wavetable is currently being played back: as a performer moves their arm, they are able to control and discern slight timbral shifts in the sonic output. Similarly, the pitch bend of the kick is also controlled by the gyroscopic data, bending forward or upward resulting in lower or higher pitches.

The Pi is also equipped with a Sparkfun Qwiic Micropressure sensor, which detects the internal pressure of a plastic balloon attached to it with a clear tube: the performer can squeeze

the balloon and create a pressure fluctuation. This data is used to control the pitch bend of the drone layer and the delay mix of the kick (both are linear mappings).

To be able to hear the audio outputs, a Freenove DAC is connected to the board and provides audio input to a PAM8610 Amp Board using a 3.5mm audio jack. The amp is connected to a speaker and a transducer.

The gesture to sound mapping strategy for this project was achieved through a mix of metaphors and experimentation. Firstly, certain gestures can evoke certain sounds (notably, pitch is often understood in terms of verticality). As such, we spend time considering which gestures could have sonic analogies. For instance, scanning through a wavetable via rotating of the wrist seemed effective, since it can evoke switching through channels with a knob, looking through a collection of different yet related sounds. From these initial metaphors, we fine-tuned the amount of modification a certain parameter would have by testing different ratios and coefficients of our raw data.

One important major change between the prototype and Psismos is the synthesis engine. While the prototype used Max/MSP on a separate PC, the final uses synthio, which runs on the Pi. This decision was partly made due to the fact that using Max/MSP would either require the performer to be connected to the PC via microUSB to send data, or implement a networked communication between the board and the PC. Both options represented drawbacks (mobility vs complexity) and synthio helped curb both of these problems. While the synthesis may not be as flexible as in Max, it still serves its purpose and provides interesting outputs.

An important formal addition to the sleeve from its prototype shape was threading along the side seams, which created visible tension and allowed for more flexibility and adjustment in the material. The right side seam can be tightened to accommodate different forearm sizes, resembling a corset design.

An acrylic cylinder is placed on top of the speaker, with a transparent balloon sitting inside. The user/performer is mostly obscured while the balloon is brightly illuminated by a white light enveloped by rope. The balloon is sitting in a synopticon-style device while the performer is moving behind it, controlling the sound, obscured.

A rope harness holds the transducer tightly to the user’s chest and they are subjected to the vibrations of the device as well as the compression of the harness.

Observations

What works

In general, this final iteration of our project fulfills the intended experience and design goals. Both sensors work in tandem to capture different data generated by a performer’s body and sound is used to make the user feel and hear the effects of their movement. A highlight of this project is the haptic feedback, because it offers users a novel and unexpected way of interacting with sound. Having a wearable sleeve as a controller is also effective at removing the possible

interaction barrier a more traditional-looking controller might represent: anyone who can move their arms and squeeze a ball can explore this piece without prior musical experience. The sound design also effectively communicates the intended context we envisioned.

The cylinder, balloon and light were very visually effective. They shifted the attention from the performer and onto the environment. This echoes an aspect of the rave experience, where the environment becomes the observed element and the ravers are dissolved into a crowd, permitting a release of inhibitions and connection to the sound & self.

Finally, while the relevance of the rope design for the chest piece was questioned during the final Q&A, we believe its visual connotations and constricting effects of the body work well with the overall concept of the piece.

What needs work

The correlation between gesture and sound needs work. While the mappings function, they are not always as clear or expressive as intended.

The wires dangling around and the heaviness of the components impeded the movement of the user. The performance was even cut short by a power cable popping off from the amplifier. This showed how much the current form still depends on sometimes cumbersome infrastructure.

Here is a video of the final performance: https://drive.google.com/file/d/105DxOkhGMqxSAnZVvtMKViwSKoPytoPt/view?usp=sharing

Future Directions

Multiple directions emerged for future development. Firstly, the sound mapping could be further refined so that the relationship between gesture, pressure, and sonic change becomes more intuitively legible.

Second, the system should be completely portable. Reducing cable dependence and redistributing weight would allow a greater range of movement.

Third, the project could be expanded toward crowd participation, like Erin Gee's work (namely *Pinch and Soothe*) or other participatory performance works. Eventually, it could be interesting to explore a wider range of gestures and eventually curate them into a performance score given to participants interacting with the piece for the first time. Feedback during the Q&A also raised the possibility of participatory theatre or a more collective performance format. This direction is aligned with our interest in rave culture, communal embodiment and the relationship between individual sensation and shared experience.

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